

LOGISTIC REGRESSION (SCRATCH)

Sigmoid function $\rightarrow$ predicts probability range.

Formula.

$$1 / (1 + e^{-z})$$

$$z = mx + b$$

Evaluate the

$\rightarrow$ using Accuracy, Confusion Matrix & Classification Report

in algo $\rightarrow$ weight & bias values

Reason: if bias value $\rightarrow 0$ it will multiply with it & discard it

So assign some weight initially.

eg bias = 2.

$\Rightarrow$ Gradient Descent: It will help in giving priority of x_0, x_1, x_2 and assigning weights.

$\Rightarrow$ hyperparameters: Before training $\rightarrow$ parameters.

$\Rightarrow$ $\rightarrow$ Parameters: During training $\rightarrow$ parameters are changing.

def sigmoid(self, z)

any real number z as input

① e = Euler's number

② value b/w 0 & 1

$dw = \frac{1}{n}$ weights

Tells how to adjust each feature's weight

def fit $\rightarrow$ Training.

① Trains the Logistic Regression Model using Gradient Descent

$db = \frac{1}{n}$ bias

① Tells how to shift the whole prediction up or down.

Polynomial Regression

⊙ for discontinuous data

like 500, 100, 100 (it will not be a straight line).

⊙ Used when data has a curve
Not a straight line

⊙ `poly = PolynomialFeatures (degree = 4)`

`X - poly = poly . fit - transform (X)`

↳ expands X into $[1, x, x^2, x^3, x^4]$

⊙ Train using Linear Regression.